

Polynomials, Part III: Graphs of Polynomial Functions and Finding Polynomials Given Their Zeros

Write a polynomial function in standard form that has the following zeros, leading coefficient (LC), and degree (D).

① $-2, -4, 5$ $LC=4$ $D=3$

② $1, 2, 3, 4$ $LC=1$ $D=4$

③ $i, -3$ $LC=5$ $D=3$

④ $2-2i, -1, 1$ $LC=1$ $D=4$

$$\textcircled{5} \quad -3i, -5, 10 \quad LC = -1 \quad D = 4$$

Indicate the end behavior for each function.

$$\textcircled{6} \quad f(x) = 6x^4 - x^3 + 2x - 11$$

End Behavior:

$$\textcircled{7} \quad f(x) = -x^2 + 20x + 16$$

End Behavior:

$$\textcircled{8} \quad f(x) = 5x^5 - 7x^4 + 2x^3 - 9x^2 + 4x - 2$$

End Behavior:

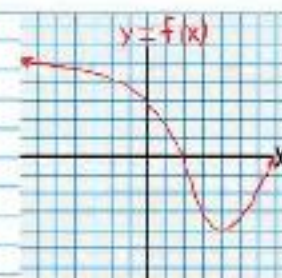
$$\textcircled{9} \quad f(x) = -8x^3 - 17x^2 - 18x - 3$$

End Behavior:

The graphs of various functions are written below. Write the intervals where each function is increasing or decreasing.

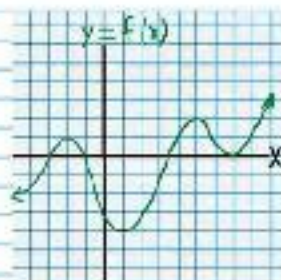
• $\textcircled{10}$ Increasing:

Decreasing:



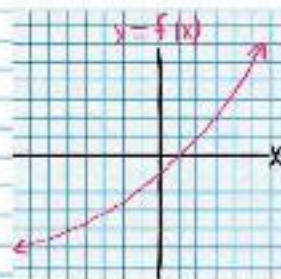
⑪ Increasing:

Decreasing:



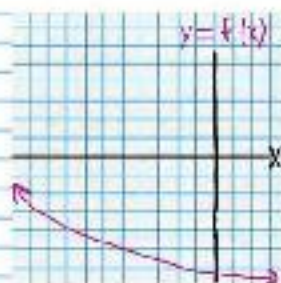
⑫ Increasing:

Decreasing:



⑬ Increasing:

Decreasing:



For each polynomial function, state the maximum number of turning points of the corresponding graph.

⑭ $f(x) = -x^3 + 20x + 16$

Max # of Turning Points:

⑮ $f(x) = 5x^5 - 7x^4 + 3x^3 - 9x^2 + 4x - 2$

Max # of Turning Points:

⑯ $f(x) = -x^2 - x - 2$

Max # of Turning Points:

Polynomial functions are written below in factored form. Write the x intercepts of each corresponding graph and determine if the function crosses or touches the x -axis at each intercept.

⑰ $f(x) = -(x-5)^2(x-\sqrt{5})(x+1)^2$

$$(18) f(x) = 3(x+10)^5(x+3)^9$$

$$(19) f(x) = (x-2\pi)^{50}(x+17)^{31}(x+8.64)^{28}(x-102)^{21}$$

Answers

$$(1) f(x) = 4x^5 + 4x^2 - 88x - 160$$

$$(2) f(x) = x^4 - 10x^3 + 35x^2 - 50x + 24$$

$$(3) f(x) = 5x^3 + 15x^2 + 5x + 15$$

$$(4) f(x) = x^4 - 4x^3 + 7x^2 + 4x - 8$$

$$(5) f(x) = -x^4 + 9x^3 + 41x^2 + 45x + 150$$

$$(6) \text{ As } x \rightarrow \infty, f(x) \rightarrow \infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow \infty$$

$$(7) \text{ As } x \rightarrow \infty, f(x) \rightarrow -\infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty$$

$$(8) \text{ As } x \rightarrow \infty, f(x) \rightarrow \infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty$$

$$(9) \text{ As } x \rightarrow \infty, f(x) \rightarrow -\infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow \infty$$

$$(10) \text{ Increasing: } (4, 7)$$

$$\text{Decreasing: } (-7, 4)$$

$$(11) \text{ Increasing: } (-5, -2) \text{ and } (1, 5) \text{ and } (7, 9)$$

$$\text{Decreasing: } (-2, 1) \text{ and } (5, 7)$$

$$(12) \text{ Increasing: } (-8, 6)$$

$$\text{Decreasing: } x$$

$$(13) \text{ Increasing: } x$$

$$\text{Decreasing: } (-11, 3)$$

$$(14) 1$$

$$(15) 4$$

$$(16) 6$$

$$(17) 5 \Rightarrow \text{Touches}$$

$$\sqrt{6} \Rightarrow \text{Touches}$$

$$-1 \Rightarrow \text{Touches}$$

$$(18) -10 \Rightarrow \text{Crosses}$$

$$-3 \Rightarrow \text{Crosses}$$

$$(19) 2\pi \Rightarrow \text{Touches}$$

$$-17 \Rightarrow \text{Crosses}$$

$$2.64 \Rightarrow \text{Touches}$$

$$102 \Rightarrow \text{Crosses}$$